



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
FEBRUARY 2023**

Course : EC3114 (Software Modelling and Analysis) Time : 8.00 am – 11.00 am
(3 hours)

Lecturer : Noor Zuhaili bt Md Yasin

Date : 18 May 2023

Instructions:

Answer **ALL** questions.

The maximum number of marks is 100.

This question paper consists of 3 printed pages
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. What is a requirement engineering? Explain your answer. (4 MARKS)
 - b. List **THREE (3)** methods of gathering requirements. (3 MARKS)
 - c. With suitable examples, explain **TWO (2)** Functional Requirements and **TWO (2)** Non-Functional Requirements. (8 MARKS)
 - d. Describe feasibility study in requirement engineering. (5 MARKS)
 - e. What is risk analysis in requirement engineering? Give **TWO (2)** examples of risk factors. (5 MARKS)
2.
 - a. Describe the process of server delivering services to clients in client-server architecture. (4 MARKS)
 - b. Explain **THREE (3)** benefits of Unified Modeling Language (UML) in modeling. (9 MARKS)
 - c. Draw a collaboration diagram for the following scenario: (12 MARKS)

System ABC prompts the username and password when the user clicks on a login button to check that the user has authorization and logs the user in. If there is a problem with the name or password, the user is given one more chance to enter a valid name and password, otherwise the login program goes into a 5-minute time-out, then returns to idle mode. While the user is logged in, the user is allowed to execute different applications by clicking on appropriate buttons. The session terminates when the user clicks the log out button.
3. To take an exam, an instructor first notifies the students of the exam date and the material to be covered. She then prepares the exam paper (with sample solutions), gets it

copied to produce enough copies for the class, and hands it out to students at the designated time and location. The students write their answers to exam questions and hand in their papers to the instructor. The instructor then gives the exam papers to the TAs, along with sample solutions to each question, and gets them to mark it. She then records all marks and returns the papers to the students.

a. Draw a sequence diagram representing this process. (10 MARKS)

b. Draw a use case diagram for the above scenario. (5 MARKS)

c. Draw a state machine diagram to show the following processes: (10 MARKS)

- To pass the course, the student must attend two lectures, present to the class a paper she read, and complete a course project, due on the last day of the term.

- To give a presentation, the student is given a date by the instructor, prepares the presentation, and gives it on the assigned day. At any time, the student can drop the course.

4. Consider the world of libraries. A library has books, videos, and CDs that it loans to its users. All library material has an id# and a title. In addition, books have one or more authors, videos have one producer and one or more actors, while CDs have one or more entertainers. The library maintains one or more copies of each library item (book, video, or CD). Copies of all library material can be loaned to users. Reference-only material is loaned for 2 hours and can't be removed from the library. Other material can be loaned for 2 weeks. For every loan, the library records the user, the loan date and time, and the return date and time. For users, the library maintains their name, address, and phone number.

- a. Draw a class diagram for the description above. Make sure to show attributes, multiplicities and aggregations or compositions where appropriate. **(15 MARKS)**
- b. Identify **FIVE (5)** functional requirements from the above case study. **(10 MARKS)**

-END OF QUESTION PAPER-